Assumptions, game formats, and configurations

A reproducibility note for the PG results. The formats below are complementary conditions; neither is designated the primary game. This note covers an exploratory eight-seed comparison, not a convergence or publication-readiness claim.

1. Environment and game definitions

Condition	Rules actually used	Outcome reported
First goal · deterministic A10	7×5; goal rows y=1–3; simultaneous U/D/L/R; deterministic A10 collision resolution; first goal ends play. For the matched critic comparison, A10 kickoff is P0 (0,1), P1 (6,3), ball P0.	Who scores first; a 100-step cap handles no-goal ties.
First goal · random move order	Same board dimensions, goal rows, actions, first-goal termination and 100-step cap. The transition randomly chooses which player resolves first when order can affect the joint move. This is a stochastic transition rule, not a random board layout. Historical PG runs use SoccerGame defaults for kickoff: (0,2) and (6,2), ball P0.	Who scores first. Seat-specific rates are kept separate because kickoff value is asymmetric.
Fixed horizon · goal reset	Deterministic move order; after a goal, players return to their start cells and the conceding player receives the ball. Play continues to 100 steps. Historical rate runs use SoccerGame defaults for kickoff (0,2) and (6,2), not A10SoccerGame's ID-specific kickoff.	Total goal difference over 100 steps; ties are equal goal counts.

Shared payoff and solver conventions

The game is two-player zero-sum. Player 0 attacks right and player 1 attacks left. A goal gives the scorer +1 and the other player −1; non-goal steps give 0. The discount factor is γ=0.9 for these PG comparisons. No reward shaping or bonus is used. The exact finite-horizon solver receives the same transition/reward definition and remaining time as the learned policy.

Important distinction: "goal-reset A10" is a historical report label for the deterministic 7×5 goal-reset variant. The implementation constructs SoccerGame(scoring="rate"), whose default kickoff differs from the assignment-specific A10SoccerGame kickoff. Interpret those results with that kickoff assumption.

Assumptions and scope limits

Assumption	How it is treated
Zero-sum objective	The score is P0's payoff and P1's is its negative; the exact reference and Nash gap use that objective.
Seat handling	P0 and P1 are evaluated separately. A 50:50 seat mean is the arithmetic mean of seat-specific empirical win rates, not a claim that the seats are strategically identical.
Finite horizon	All neural policies observe remaining steps. The 100-step cutoff is part of the game; first-goal episodes can terminate earlier.
No symmetry constraint	Players use independent networks. Any mirror relation is measured, not imposed.

Observed eight-seed summary

Method	Mean Nash gap (range)	P0 win vs. BR	P1 win vs. BR
A2C 10-step	0.865 (0.839–0.898)	2.1%	0.9%
A2C learned MC baseline	0.849 (0.748–0.940)	1.5%	0.6%
A2C frozen exact-value baseline	0.874 (0.808–0.928)	0.6%	0.7%

Frozen exact-value minus learned-MC gap: +0.025 payoff units; 95% paired t interval −0.038 to +0.089, across eight seed labels. Frozen has a higher gap on four seeds and a lower gap on four. The interval spans zero; this small exploratory comparison provides no reliable evidence of improvement.

What was held fixed, and what the metrics mean

2. Matched eight-seed critic comparison

Setting	Value used
Game / format	Deterministic A10; first-goal termination; 7×5, goal rows 1–3, A10 kickoff (0,1)/(6,3), ball P0; U/D/L/R; 100-step cap; γ=0.9.
Seeds and budget	Seeds 4–11 (eight runs per arm); 2,000 iterations; 64 episodes per collected batch; 100 steps maximum per episode.
Optimizer cadence	Five shuffled full-batch update passes for REINFORCE/A2C arms in this comparison. Each policy network is 64×64; learning rate 0.001; entropy coefficient 0.01. PPO is not part of this three-arm critic comparison.
Fictitious play	Snapshot every 5 iterations. Each learner samples an opponent snapshot uniformly from that opponent's saved history; the reported own policy averages snapshot action probabilities, not network weights.
Compared arms	A2C 10-step learned critic (context); A2C with an online Monte Carlo value baseline; A2C with a state-time value network fitted to exact solver V values for 10 epochs then frozen. Both critic-comparison actors use sampled full discounted returns minus their baseline. The actor does not query exact Q values.
Evaluation	Exact finite-horizon best responses; 1,000 sampled games per player seat and seed. Win-rate opponents best-respond to discounted score difference.

3. Earlier PG benchmark settings

The concise PG report also summarizes the repository's earlier three-seed comparisons: REINFORCE, A2C, and PPO in standard self-play and neural fictitious play; first-goal deterministic A10, first-goal random move order, and deterministic goal-reset. The shared sweep baseline is 2,000 iterations, 64 episodes per batch, 100-step maximum horizon, γ=0.9, learning rate 0.001, entropy 0.01, and separate 64×64 player networks. Saved REINFORCE/A2C runs used one update per collected batch; PPO used four epochs. Fictitious-play snapshots were every five iterations. Variants such as trimming, replay, minibatches, alternate snapshot intervals, and entropy changes were separate ablations, not pooled into the benchmark table.

Historical results are rounded summaries from the earlier report and are not pooled with seeds 4–11. For exact new per-run data, use experiments/pg_paired_new_seeds_a10.csv and scripts/pg_paired_seed_extension.py. The earlier three-seed summaries do not support a controlled cross-format causal comparison.

4. Reading the metrics without overclaiming

Metric	Definition and limit
Win rate	Fraction of sampled games won from a given seat. It is an outcome frequency. The exact opponent minimizes expected discounted score difference, not win probability; therefore win rate is not the optimization target.
Nash gap / exploitability	Sum of both unilateral gains in expected discounted score when each player replaces its policy with an exact best response to the other. Units are payoff; zero means equilibrium for this finite-horizon solver. Smaller is closer.
Seed comparison	The same seed labels are used across arms, but action trajectories are not identical: policies diverge, stochastic episodes differ, and value pretraining consumes random draws. Seeds 7–9 extended the original pilot; seeds 10–11 were selected after reviewing that pilot. None is a preregistered holdout, so the eight-seed extension is exploratory.

Recent theory does not transfer automatically. Independent policy-gradient guarantees require the paper's two-timescale update rule and assumptions; the current fixed-rate fictitious-play comparison does not implement that method. The 2025 convex-Markov-game result uses a distinct regularized game formulation.

Sources: `scripts/pg_finite.py`, `scripts/pg_report.py`, `soccer_nash/game.py`, `soccer_nash/finite_horizon.py`, and `experiments/pg_paired_new_seeds_a10.csv`. Theory references: Daskalakis et al. (NeurIPS 2020); Kalogiannis et al. (ICML 2025).

No convergence, novelty, or publication-readiness claim is made by this assumptions note.